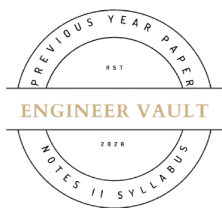


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**SUBJECT CODE NO: RNP- 8001**  
**FACULTY OF SCIENCE AND TECHNOLOGY**  
**F.E (Group A & B) (NEP) Sem - I**  
**Examination May/June-2026**  
**Applied Mathematics-I**

**[Time: 3:00 Hours]****[Max. Marks: 60]**

Please check whether you have got the right question paper.

N. B

- 1) Use of non-programmable calculator is allowed
- 2) Q. no.1 is compulsory
- 3) Solve any four questions from Q. nos 2, 3, 4, 5, 6 and 7

**Q1 Solve any Ten of the following questions****20**

1) By comparison test "If two positive terms  $\sum u_n$  &  $\sum v_n$  be s. t.  $\lim_{n \rightarrow \infty} \frac{u_n}{v_n} = k$  "then

- a)  $\sum u_n$  series is convergent or divergent
- b)  $\sum v_n$  series is convergent or divergent
- c)  $\sum u_n$  &  $\sum v_n$  both series is convergent or divergent
- d) None of the above

2) The value of  $\lim_{x \rightarrow 1} \left( \frac{1}{\log x} - \frac{x}{x-1} \right)$  is

- a)  $\frac{1}{4}$       b) 0      c) -1      d)  $-\frac{1}{2}$

3) The series expansion  $x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots$  is of

- a)  $\cos x$     b)  $\tan x$     c)  $\sin x$     d)  $e^x$

4) Find the Taylor series expansion of the function  $\cos h(x)$  at  $x = 0$ .

- a)  $1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots \infty$
- b)  $\frac{x}{1!} + \frac{x^3}{3!} + \frac{x^5}{5!} + \dots \infty$
- c)  $1 + \frac{x^2}{2!} + \frac{x^4}{4!} + \dots \infty$
- d)  $1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots \infty$

5) If  $u$  is homogeneous function of  $x, y$  of degree  $n$ , then by Euler's thm:  $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y}$  is equal to

- a) 0      b) 1      c)  $nu$       d)  $u$

6) If  $u = \log(x^2 + y^2)$  then  $\frac{\partial u}{\partial x}$  is

- a)  $\frac{3x+3y}{x^3y^3}$       b)  $\frac{2x}{x^2+y^2}$       c)  $\frac{3y}{x^3y^3}$       d)  $\frac{3y^2}{x^3+y^3}$

7) Divide 120 into three parts so that the sum of their products taken two at a time

shall be maximum if

a)  $x = 40, y = 40, z = 40$

b)  $x = 30, y = 30, z = 60$

c)  $x = 30, y = 45, z = 45$

d)  $x = 40, y = 30, z = 50$

8) If  $u, v$  &  $w$  are functions of three independent variables  $x, y$  &  $z$  then the Jacobean is given by

a)  $J = \frac{\partial(x,y,z)}{\partial(u,v,w)}$

b)  $J = \frac{\partial(x,v,z)}{\partial(u,y,w)}$

c)  $J = \frac{\partial(u,v,w)}{\partial(x,y,z)}$

d)  $J = \frac{\partial(z,y,x)}{\partial(w,y,x)}$

9) The necessary & sufficient condition for D.E. to be exact D.E. is

a)  $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$

b)  $\frac{\partial M}{\partial y} = -\frac{\partial N}{\partial x}$

c)  $\frac{\partial M}{\partial x} = \frac{\partial N}{\partial x}$

d)  $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial y}$

10) Newton's second law of motion is

a)  $F = m \frac{dv}{dt}$

b)  $F = v \frac{dv}{dx}$

c)  $F = \frac{dv}{dx}$

d)  $F = ma \frac{dv}{dx}$

11) The stationary points of the function  $f(x, y) = x^3 y^2 (1 - x - y)$  are origin

a)  $x = 0, y = 1/3$

b)  $x = 1/2, y = 0$

c)  $x = 1/2, y = 1/3$

d)  $x = 1/2, y = -1/3$

12) A circuit containing resistance  $R$  and inductance  $L$  in series with voltage source  $E$  then the D.E for current is

a)  $Li + R \frac{di}{dt} = E$

b)  $L \frac{di}{dt} + Ri = E$

c)  $Li - R \frac{di}{dt} = E$

d)  $Li + R \frac{di}{dt} = 0$

13) The Integrating Factor of  $\frac{dy}{dx} + xy = x^3$  D.E is

a)  $x^2 y = c$

b)  $x = c$

c)  $e^{\frac{x^2}{2}}$

d)  $x^2 y - xy = c$

Q.2

a) Evaluate:  $\lim_{x \rightarrow 0} \left[ \frac{a}{x} - \cot \frac{x}{a} \right]$

05

b) Test the convergence of series  $\sum \frac{n^2(n+1)^2}{n!}$

05

Q.3

a) Solve:  $\sin x \frac{dy}{dx} + y \cos x = 2 \sin^2 x$ .

05

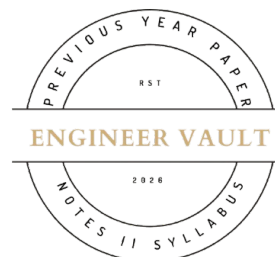
b) Solve:  $\cos x \frac{dy}{dx} = -\frac{4x^3 y^2 + y \cos xy}{2x^4 y + x \cos xy}$

05

Q.4

a) If  $u = \sin^{-1} \frac{x+y}{\sqrt{x}+\sqrt{y}}$  then find  $x^2 u_{xx} + 2xy u_{xy} + y^2 u_{yy}$

05



b) If  $z = y^x$  then show that  $\frac{\partial^2 z}{\partial x \partial y} = \frac{\partial^2 z}{\partial y \partial x}$

05

**Q.5** a) Expand  $\log \sec x$

05

b) Use Taylor's theorem to find  $\sqrt{10}$

05

**Q.6** a) If  $xu = vw, yv = wu, zw = uv$  then prove that  $\frac{\partial(x,y,z)}{\partial(u,v,w)}$

05

b) A rectangular box open at the top is to have volume 256 cubic feet, determine dimensions of the box required least amount of material.

05

**Q.7** a) A resistance of  $100\Omega$ , an inductance of 0.5 Henry is connected in series with a battery of 20 volts. Find the current in the circuit at  $t = 0.5$  sec if  $i = 0$  at  $t = 0$

05

b) A body of mass  $m$ , falls from rest under gravity in a fluid whose resistance to motion at any instant is  $mK$  times its velocity, where  $K$  is a constant. Find the terminal velocity of the body & also the time to acquire one half of its limiting speed.

05

